

The Watermark Law

Discriminant laws for the linear Hodge lattices of Fermat surfaces: one exponent formula for every degree, five blind verdicts, and an off-by-one in a published table

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4 June 2026

Abstract

Reading the elementary-divisor tables of Aljovin–Movasati–Villaflor [3] jointly with the line-lattice results of Schütt–Shioda–van Luijk [2] and Shioda’s 1987 conjecture [1], we state a single exponent law for the discriminant of the lattice V of linear Hodge cycles of the complex Fermat surface of every degree $m \geq 3$: $|\text{disc } V| = m^{3(m-3)^2}$ for m odd, with the extra factor $(m/4)^3$ for m even. The odd prime case is Shioda’s conjecture; the odd verification to degree 81 is [2]; *the even half is stated in no source we could find*. The law fits all twelve rows of [3, Table 1] and was then tested by five sealed blind predictions — seven exponents, seven exact hits — three of the cells in even degree, outside every swept source. We also report a finer law for the divisor profiles of odd prime degree, a corner law for when $\ell(\text{disc})$ meets rank T , and one off-by-one in the published $(6, 4)$ row of [3, Table 2], caught by an independently measured rank. All claims carry explicit confidence flags; every number is reproducible from the public repository.

1 The unified law

Let $V = V_m^2$ be the lattice of linear Hodge cycles of [3] for the Fermat surface of degree m (full register, their Table 1); for $m \leq 4$ or $\gcd(m, 6) = 1$ it is the full Néron–Severi lattice.

Law (the Watermark law; CONJ, blind-tested). $|\text{disc } V(2, m)| = m^{3(m-3)^2}$ for m odd, and $|\text{disc } V(2, m)| = m^{3(m-3)^2} \cdot (m/4)^3$ for m even.

Verified on all twelve rows of [3, Table 1] ($m = 3, \dots, 14$), prime powers included with no further correction ($m = 8 \mapsto 2^{228}$, $m = 12 \mapsto 2^{486} 3^{246}$). For $m = 2q$, q an odd prime, the factor $(q/2)^3$ splits as -3 on the prime 2 and $+3$ on q : $\exp_2 = 12(q-1)(q-2)$ and $\exp_q = \exp_2 + 6$. We refer to the statement above as the *Watermark law*: the degree stamps its watermark on the discriminant, and the law reads it off the degree alone — no lattice computation required.

Five blind verdicts. All predictions sealed in writing on 4 June 2026 before any computation; instrument and gates in §4. Ranks are $3(m-1)(m-2) + 1$ (m odd) and $3(m-1)(m-2) + 2$ (m even), per [2, eq. (3)].

cell	lines	rank	sealed	measured	verdict
(2, 17)	867	721	$\text{exp}_{17} = 588$	588	PASS
(2, 19)	1083	919	$\text{exp}_{19} = 768$	768	PASS
(2, 16)	768	632	$\text{exp}_2 = 2034$	2034	PASS
(2, 18)	972	818	$\text{exp}_2 = 672, \text{exp}_3 = 1356$	672, 1356	PASS
(2, 20)	1200	1028	$\text{exp}_2 = 1734, \text{exp}_5 = 870$	1734, 870	PASS

The odd cells sit inside the determinant range verified by [2] (instrument validation); the three even cells lie outside every swept source and blind-test the law’s new half. Every run was reproduced by a second independent execution before being recorded.

2 The profile law, the ceiling corner, the erratum

Profiles. For odd prime $m \geq 7$: $\#(m^2\text{-divisors}) = 3m - 16$ and $\#(m\text{-divisors}) = 3m^2 - 24m + 59$, all heights ≤ 2 — five of five on $m = 7, 11, 13, 17, 19$, the last two computed here for the first time ($17^{518} \cdot 289^{35}$ and $19^{686} \cdot 361^{41}$). The named exception: $m = 5$ has height-2 count $1 \neq -1$; the checksum $h_1 + 2h_2 = 3(m - 3)^2$ holds at all six points. No swept source publishes these profiles ([2] give determinants only; [3] stop at $m = 14$).

Ceiling corner [FIRM on 19 clean points]. With ℓ the number of nontrivial divisors and T the transcendental lattice (recall $\text{disc } V \cong \text{disc } T$ by saturation in the unimodular H^2): across [3, Table 1] and the three uncontaminated rows of their Table 2, $\ell = \text{rank } T$ exactly iff $n \leq 4$ and $d \leq 4$. The first guess “iff $d \leq 4$ ” is refuted by the clean $d = 3$ column: slack 1, 11, 78 at $(6, 3), (8, 3), (10, 3)$ — a gradient recorded as an open structure question; cyclotomic-field explanations die with it (same $\mathbb{Q}(\zeta_3)$).

The erratum [FIRM]. The $(6, 4)$ row of [3, Table 2] prints $-1^{576} 2^{68} 4^{49} 8^{296} 16^{48} 32^{69} 512^2$, summing to rank 1108. The rank is 1107: it equals $\text{DS}_6(4)$ of [4, Remark 4.4], independently measured on hardware by this campaign, and equal to the central trinomial walk count $T(8) = 1107$. The row-sum checksum passes 3-of-3 on the table’s other rows (70, 252, 924 — the last against this campaign’s own measured $(10, 3)$), so exactly one printed exponent is off by one; which one, only the authors’ raw data can say. No theorem of [3] is affected.

3 Ownership

Odd prime m , exponent conjectured for NS: Shioda 1987 [1, Questions 7.2, 7.4(i), p. 133]. Odd $m \leq 81$ verified: [2, Remark 4.5, eq. (16)]. Mechanism for prime m (m^3 per Galois orbit, $(m - 3)^2$ orbits, square part = Brauer): [1, §§6–7]. **Even m : stated nowhere** — [1] is prime-only (general m receives divisibility only, Remark 6.4 and Added in proof); [2] exclude even m from their basis machinery (Remark 4.3(iii): $\omega = -1$ unavailable); absent from [3] and from Degtyarev’s survey arXiv:1512.06199. Doubly so: Shioda’s question concerns NS, and for even $m \geq 6$ the lines do not rationally generate NS [2, Thm. 3.1] — the even law lives on V . The route to a theorem is in [1] itself: the prime-only ingredient is Iwasawa’s congruence $j(\alpha) \equiv 1 \pmod{(1 - \zeta_m)^3}$; the even correction $(m/4)^3 = m^3/2^6$ is a candidate 2-adic defect of exactly that congruence, to be combined with the refined Artin–Tate formula of [1, §6].

4 Instrument and protocol

`GRAM_MANOMETER_SURFACE_v4.py` (repository): Gram matrix of the $3m^2$ lines from the intersection rules (5) of [2] — self-intersection $2 - m$; for even m the (1,3) rule carries ω^2 explicitly — and layered p -adic elimination with a shrinking working modulus. Gates byte-exact against [3, Table 1] before every shot: (2, 5), (2, 7), (2, 6), (2, 8); for odd m the general rule reproduces the $\omega = -1$ instrument exactly, so the odd gates double as an invariance test. One self-caught precision bug (v2→v3) is part of the public record; gates were re-passed before any sealed shot. Register caveat, load-bearing: “lines generate V ” is [3, Theorem 1], proven to $m \leq 14$; at $m = 16, 18, 20$ the blind passes are double evidence (the law and the integral generation both extend).

5 What is not claimed

No progress on any Hodge-type conjecture: these lattices are torsion-free, and the laws concern their discriminants. The unified law is a conjecture with a 17-point record (12 in-sample, 5 blind); the profile law, 5 points and one named exception; the ceiling corner, a verified finite pattern; the erratum, a checksum fact.

References

- [1] T. Shioda, *Some observations on Jacobi sums*, Adv. Stud. Pure Math. **12** (1987), 119–135.
- [2] M. Schütt, T. Shioda, R. van Luijk, *Lines on Fermat surfaces*, J. Number Theory **130** (2010), 1939–1963.
- [3] E. Aljovin, H. Movasati, R. Villaflor, *Integral Hodge conjecture for Fermat varieties*, J. Symb. Comput. **95** (2019), 177–184.
- [4] A. Degtyarev, I. Shimada, *On the topology of projective subspaces in complex Fermat varieties*, J. Math. Soc. Japan **68** (2016), 975–996.
- [5] A. Degtyarev, *Lines generate the Picard groups of certain Fermat surfaces*, J. Number Theory **147** (2015), 454–477.